

PHYS 427: Lecture # 6

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Lecture Span

1. Equipartition Theorem

Recap:

We got to a one particle in a box (low and high temp limits). In the high temp limit, replace sum with integral. We get a Gaussian integral for Z . As

$$Z(T) \approx \int_0^\infty dn_x \exp(-\alpha_x n_x^2) \int_0^\infty dn_y \exp(-\alpha_y n_y^2) \dots \quad (1)$$

Such that

$$\alpha_x = \frac{\hbar^2 \pi^2}{2mL_x^2 k_B T} \quad (2)$$

The integral is then

$$Z_1(T) = \frac{\pi^{3/2}}{2^3 (\alpha_x \alpha_y \alpha_z)^{1/2}} \quad (3)$$

You can also plug in α ,

$$Z_1(T) = \left(\frac{mk_B T}{2\pi \hbar^2} \right)^{3/2} V \quad (4)$$

For $\alpha_i \ll 1$. Therefore, $k_B T \gg \hbar^2 \pi^2 / 2mL_x^2$ (same for other directions).

All above work is in **the high temperature limit**.

$$Z_{tot} = \frac{Z_1^N}{N!} \quad (5)$$

Here, use the previously calculated Z_1 . Calculate U for an ideal gas, then start at the partition function:

Note that $Z_1 \sim (\beta^{-3/2})^N$

$$U = -\partial_\beta \ln Z_{tot} = -N \partial_\beta \ln \left(\beta^{-3/2} + \text{terms independent of } \beta \right) \quad (6)$$

$$= \frac{3}{2} N \frac{1}{\beta} = \frac{3}{2} N k_B T = U \quad (7)$$

The de-Broglie wavelength

$$\lambda = \frac{h}{p} \iff p = h/\lambda$$

Then,

$$\langle \epsilon \rangle = \langle p^2 / 2m \rangle = \frac{3}{2} k_B T \quad (8)$$

Then

$$\langle \lambda \rangle = \frac{h}{\sqrt{2m\langle e \rangle}} = \sqrt{\frac{4\pi^2\hbar^2}{3mk_BT}} \quad (9)$$

Quantum Density

$$n_Q = \frac{1}{\sqrt{2\pi\hbar^2/mk_BT}^{3/2}} \quad (10)$$

For a cube, the edge length

$$\sqrt{\frac{2\pi\hbar^2}{mk_BT}} \approx \lambda \cdot \mathcal{O}(1) \quad (11)$$

$$n_Q \approx \frac{\text{constant of order 1}}{\langle \lambda \rangle^3} \quad (12)$$

$$Z_1(T) = n_Q(T)V \quad (13)$$

Consider Helium at room temperature. That is $T \approx 300\text{K}$ and atmospheric pressure. Then,

$$n_Q(T) \approx 0.8 \times 10^{25} \frac{1}{\text{cm}^3} \quad (14)$$

This is about 1 atom per 0.5 Angstrom cubed. The typical density is

$$n_{He} = 2.5 \times 10^{19} \frac{1}{\text{cm}^3} = \frac{N}{V} \quad (15)$$

This is about 1 atom per 34 Angstrom cubed. Therefore,

$$n_{He} \ll n_Q \quad (16)$$

This is the classical regime.

Equipartition Theorem

Considering quadratic degrees of freedom. This means

$$\frac{p^2}{2m} = \sum_i \frac{p_i^2}{2m} \quad (17)$$

This is 3 degrees of freedom (also quadratic!). Then,

$$\langle \epsilon_x \rangle = \langle p_x^2/2m \rangle = \frac{1}{2}k_BT \quad (18)$$

Every degree of freedom contributes $(1/2)k_BT$ to $\langle \epsilon \rangle$ in the **high temperature limit**.

In an harmonic oscillator, there are two degrees of freedom. Since you have $1/2 \cdot kx^2$ and $p_x^2/2m$. For a solid in 3D, then

$$\langle \epsilon \rangle = 3Nk_BT \quad (19)$$

General statement at high temperatures

Consider a diatomic gas, CO. Then, we have vibrational (spring system between the two atoms) and translational energy.

The total energy of the molecule is then

$$\epsilon_{tot} = \epsilon_{trans} + \epsilon_{rot} + \epsilon_{vib} \quad (20)$$

Then,

$$Z_{tot} = \sum_{\text{all states}} e^{-\beta \epsilon_{tot}} = \sum_{\epsilon_{trans}} e^{-\beta \epsilon_{trans}} \cdot \sum \dots \quad (21)$$

We can then define Z_{trans} , Z_{rot} , etc.

Consider quantum rigid motor, angular momentum. The total angular momentum J is quantized s.t. its magnitude is

$$J = |j| = \hbar \sqrt{j(j+1)}, \quad j = 0, 1, 2, \dots \quad (22)$$

$$J_z = m_j \hbar, \quad m_j = -j, -j+1, \dots, j-1, j \quad (23)$$

$$\epsilon_{rot} = \frac{|j|^2}{2I} = \frac{j(j+1)\hbar^2}{2I} \quad (24)$$

Here, I is the moment of inertia.

$$Z_{rot} = \sum_{J, J_z} e^{-\beta \epsilon_{rot}} = \sum_{j=0}^{\infty} (2j+1) e^{-\beta \epsilon_{rot}(j+1)} \quad (25)$$

The $2j+1$ term is due to degeneracy in the energy levels. In the low temperature limit, $k_B T \ll \epsilon$, then β is large and we can approximate Z_1 using a few terms.

$$Z_{rot} = 1 + 3e^{-\beta \epsilon} + 5e^{-6\beta \epsilon} + 7e^{-12\beta \epsilon} + \text{neglect the rest} \quad (26)$$

At $T \rightarrow 0$, then $Z_{rot} \approx 1 \Rightarrow U \approx 0$. Excited states are then "frozen out" at low temperatures.

At high temperatures. Use an integral instead of a sum. Then,

$$Z_{rot} = \frac{1}{\beta \epsilon} \quad (27)$$

Then,

$$U_{rot} = -\partial_{\beta} \ln Z_{rot} = k_B T \quad (28)$$

This is for $k_B T \gg \epsilon$. This makes sense since we only had 2 degrees of freedom for a rotating spin.

Vibratioanl partition function

$$Z_{vib} = \sum_n e^{-\beta \epsilon_{vib}} \quad (29)$$

Such that

$$\epsilon_{vib} = \hbar \omega (n + 1/2), \quad n = 0, 1, 2, 3, \dots \quad (30)$$

Then,

$$Z_{vib} = \sum_n e^{-\beta \hbar \omega (n+1/2)} = e^{-\beta \hbar \omega / 2} \sum_n e^{-\beta n \hbar \omega} \quad (31)$$

This is a geometric series. Then, the series converges to:

$$\frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}} = Z_{vib} \quad (32)$$

Then,

$$\langle \epsilon \rangle = U = -\partial_\beta \ln(Z_{vib}) \quad (33)$$

$$= \hbar\omega \left(\frac{1}{e^{\beta\hbar\omega} - 1} + \frac{1}{2} \right) \quad (34)$$

$$= \hbar\omega \left(\langle n \rangle + \frac{1}{2} \right) \quad (35)$$

Then, we can deduce that

$$\langle n \rangle = \frac{1}{e^{\beta\hbar\omega} - 1} \quad (36)$$

In the low temperature limit, $k_B T \ll \hbar\omega$, then we obtain

$$\langle n \rangle \approx e^{-\beta\hbar\omega} \quad (37)$$

Then,

$$U_{vib} \approx \hbar\omega \left(e^{-\beta\hbar\omega} + \frac{1}{2} \right) \approx \frac{\hbar\omega}{2} \quad (38)$$

This is valid in the low temperature limit, and the zero point energy. In the high temperature limit, then $k_B T \gg \hbar\omega$, then

$$\langle n \rangle \approx \frac{1}{1 + \beta\hbar\omega - 1} = \frac{k_B T}{\hbar\omega} \quad (39)$$

Therefore,

$$U_{vib} \approx \hbar\omega \left(\frac{k_B T}{\hbar\omega} + \frac{1}{2} \right) = k_B T + \frac{\hbar\omega}{2} \approx k_B T \quad (40)$$

This is valid in the high temperature limit. This is consistent with the equipartition theorem (**which is only valid at the high temperature limit!**).

Note,

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V \quad (41)$$

For a diatomic gas, then they got a curve that looked like a positive step function with steps at 3/2, 5/2, etc. Then,

$$C_V^{trans} = \frac{3}{2} N k_B \quad (42)$$

$$C_V^{rot} = N k_B, \quad k_B T \gg \epsilon \quad (43)$$

$$C_V^{vib} = N k_B, \quad k_B T \gg \epsilon \quad (44)$$

Then, C_V is the sum of all the above. At lower temperatures, only C_V^{trans} mattered. At high temperatures, C_V is the sum of all the above. In between, you still have some rotational and translational, so 5/2.

Lecture 6 has the full derivation of the theorem.